Exp 6 MATLAB CODE

```
clc;
clear;
close all;

% Parameters
Fs = 100000;
t = 0:1/Fs:0.01;
fm = 1000;
fc = 10000;
Am = 1;
Ac = 1;
kf = 5000;
kp = pi/2;

% Message Signal
m = Am*cos(2*pi*fm*t);

% Frequency Modulation (FM)
int_m = cumsum(m)/Fs;
fm_signal = Ac*cos(2*pi*fc*t + 2*pi*kf*int_m);

% Phase Modulation (PM)
pm_signal = Ac*cos(2*pi*fc*t + kp*m);

% Plot Message Signal
figure;
subplot(3,1,1);
```

```
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Plot FM Signal
subplot(3,1,2);
plot(t,fm_signal);
title('Frequency Modulated (FM) Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Plot PM Signal
subplot(3,1,3);
plot(t,pm_signal);
title('Phase Modulated (PM) Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

% Frequency Spectrum Analysis
N = length(t);
f = (-N/2:N/2-1)*(Fs/N);

FM = fftshift(abs(fft(fm_signal))/N);
PM = fftshift(abs(fft(pm_signal))/N);
```

```
figure;
subplot(2,1,1);
plot(f,FM);
title('FM Frequency Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
xlim([-20000 20000]);
grid on;

subplot(2,1,2);
plot(f,PM);
title('PM Frequency Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
xlim([-20000 20000]);
grid on;
```

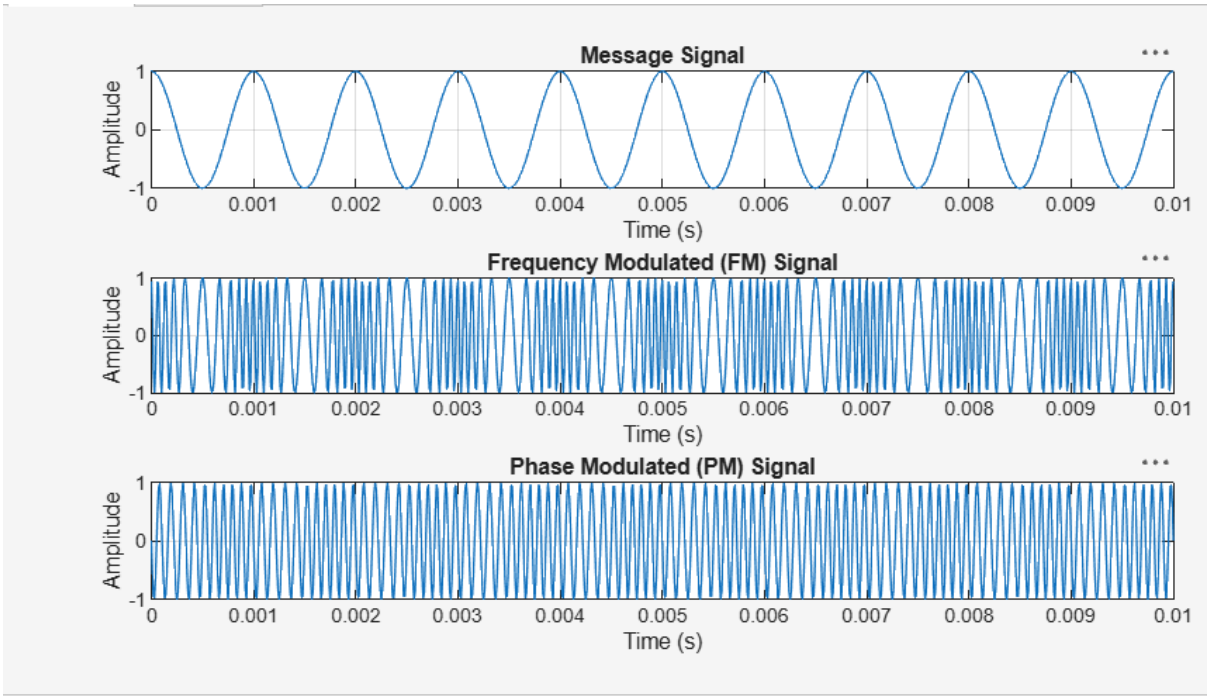


Figure 1 × Figure 2 ×

